Topic 11: Nuclear Radiation

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include measuring the half-life of a radioactive material.

Mathematical skills that could be developed in this topic include applying the concepts underlying calculus (but without requiring the explicit use of derivatives or integrals) by solving equations involving rates of change, for example. $\Delta x / \Delta t = -\lambda x$ using a graphical method or spreadsheet modelling and understanding probability in the context of radioactive decay.

This topic may be studied using applications that relate to nuclear radiation, for example nuclear power stations and medical physics.

Students should:
164. understand the concept of *nuclear binding* energy and be able to use the equation $\Delta E = c^2 \Delta m$ in calculations of nuclear mass (including mass deficit) and energy
165. use the *atomic mass unit (u)* to express small masses and convert between this and SI units
166. understand the processes of nuclear fusion and fission with reference to the binding energy per nucleon curve
167. understand the mechanism of nuclear fusion and the need for very high densities of matter and very high temperatures to bring about and maintain nuclear fusion
168. understand that there is background radiation and how to take appropriate account of it in calculations
169. understand the relationships between the nature, penetration, ionising ability and range in different materials of nuclear radiations (alpha, beta and gamma)
170. be able to write and interpret nuclear equations given the relevant particle symbols
171. **CORE PRACTICAL 15: Investigate the absorption of gamma radiation by lead.**
172. understand the spontaneous and random nature of nuclear decay
173. be able to determine the half-lives of radioactive isotopes graphically and be able to use the equations for radioactive decay: activity $A = \lambda N$, $\frac{\mathrm{d}N}{\mathrm{d}t} = -\lambda N$, $\lambda = \frac{\ln 2}{t_{1/2}}$, $N = N_0 \mathrm{e}^{-\lambda t}$ and $A = A_0 \mathrm{e}^{-\lambda t}$ and derive and use the corresponding log equations.

Topic 12: Gravitational Fields

Mathematical skills that could be developed in this topic include sketching relationships that are modelled by $y = k/x$, $y = k/x^2$.

Students should:
174. understand that a gravitational field (force field) is defined as a region where a mass experiences a force
175. understand that gravitational field strength is defined as $g = \frac{F}{m}$ and be able to use this equation
176. be able to use the equation $F = \frac{Gm_1 m_2}{r^2}$ (Newton's law of universal gravitation)
177. be able to derive and use the equation $g = \frac{Gm}{r^2}$ for the gravitational field due to a point mass
178. be able to use the equation $V_{grav} = \frac{-Gm}{r}$ for a radial gravitational field
179. be able to compare electric fields with gravitational fields
180. be able to apply Newton's laws of motion and universal gravitation to orbital motion.

Topic 13: Oscillations

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include measuring gravitational field strength using a simple pendulum and measuring a spring constant from simple harmonic motion.

Mathematical skills that could be developed in this topic include sketching relationships that are modelled by *y* = sin *x*, *y* = cos *x*.

Students should:
181. understand that the condition for simple harmonic motion is $F = -kx$, and hence understand how to identify situations in which simple harmonic motion will occur
182. be able to use the equations $a = -\omega^2 x$, $x = A\cos \omega t$, $v = -A\omega \sin \omega t$, $a = -A\omega^2 \cos \omega t$, and $T = \frac{1}{f} = \frac{2\pi}{\omega}$ and $\omega = 2\pi f$ as applied to a simple harmonic oscillator
183. be able to use equations for a simple harmonic oscillator $T = 2\pi\sqrt{\frac{m}{k}}$, and a simple pendulum $T = 2\pi\sqrt{\frac{l}{g}}$
184. be able to draw and interpret a displacement–time graph for an object oscillating and know that the gradient at a point gives the velocity at that point
185. be able to draw and interpret a velocity–time graph for an oscillating object and know that the gradient at a point gives the acceleration at that point
186. understand what is meant by *resonance*
187. **CORE PRACTICAL 16: Determine the value of an unknown mass using the resonant frequencies of the oscillation of known masses.**
188. understand how to apply conservation of energy to damped and undamped oscillating systems
189. understand the distinction between *free* and *forced oscillations*
190. understand how the amplitude of a forced oscillation changes at and around the natural frequency of a system and know, qualitatively, how damping affects resonance
191. understand how damping and the plastic deformation of ductile materials reduce the amplitude of oscillation.

Salters Horners approach

The following section shows how the course may be taught using the Salters Horners (SHAP) context-led approach. The subset of content required for the Advanced Subsidiary GCE qualification is listed on pages 27–35, while the remainder of the content required for this qualification is listed on pages 36–44.

Working as a Physicist

Throughout their study of physics at this level, students should develop their knowledge and understanding of what it means to work scientifically. They should also develop their competence in manipulating quantities and their units, including making estimates. They should experience a wide variety of practical work, giving them opportunities to develop their practical and investigative skills by planning, carrying out and evaluating experiments. Through studying a range of examples, contexts and applications of physics, students should become increasingly knowledgeable of the ways in which the scientific community, and society as a whole, use scientific ideas and methods, and how the professional scientific community functions. They should develop their abilities to communicate their knowledge and understanding of physics in ways that are appropriate to the content and to the audience.

It is not intended that this part of the specification is taught as a discrete topic. Rather, the knowledge and skills specified here should pervade the entire course and should be taught using examples and applications from throughout the rest of the specification.

Students should:	
1.	know and understand the distinction between base and derived quantities and their SI units
2.	be able to demonstrate their knowledge of practical skills and techniques for both familiar and unfamiliar experiments
3.	be able to estimate values for physical quantities and use their estimate to solve problems
4.	understand the limitations of physical measurement and apply these limitations to practical situations
5.	be able to communicate information and ideas in appropriate ways using appropriate terminology
6.	understand applications and implications of science and evaluate their associated benefits and risks
7.	understand the role of the scientific community in validating new knowledge and ensuring integrity
8.	understand the ways in which society uses science to inform decision making

Higher, Faster, Stronger (HFS)

An exploration of the physics behind a variety of sports, using use video clips, ICT and laboratory practical activities:

- graphs and equations of motion in sprinting and jogging
- work and power in weightlifting
- forces and equilibrium in rock climbing
- moments and equilibrium in gymnastics
- forces and projectiles in tennis and ski-jumping
- force and energy in bungee jumping.

There are opportunities for students to collect and analyse data using a variety of methods, and to communicate their knowledge and understanding using appropriate terminology.

Students should:	
9.	be able to use the equations for uniformly accelerated motion in one dimension: $s = \frac{(u+v)t}{2}$ $v = u + at$ $s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
10.	be able to draw and interpret displacement-time, velocity-time and acceleration-time graphs
11.	know the physical quantities derived from the slopes and areas of displacement-time, velocity-time and acceleration-time graphs, including cases of non-uniform acceleration, and understand how to use the quantities
12.	understand scalar and vector quantities, and know examples of each type of quantity and recognise vector notation
13.	be able to resolve a vector into two components at right angles to each other by drawing and by calculation
14.	be able to find the resultant of two coplanar vectors at any angle to each other by drawing, and at right angles to each other by calculation
15.	understand how to make use of the independence of vertical and horizontal motion of a projectile moving freely under gravity
16.	be able to draw and interpret free-body force diagrams to represent forces on a particle or on an extended but rigid body
17.	be able to use the equation $\sum F = ma$, and understand how to use this equation in situations where *m* is constant (Newton's second law of motion), including Newton's first law of motion where *a* = 0, objects at rest or travelling at constant velocity Use of the term *terminal velocity* is expected

Students should:	
19.	**CORE PRACTICAL 1: Determine the acceleration of a freely-falling object.**
20.	know and understand Newton's third law of motion, and know the properties of pairs of forces in an interaction between two bodies
21.	understand that momentum is defined as $p = mv$
22.	know the principle of conservation of linear momentum, understand how to relate this to Newton's laws of motion and understand how to apply this to problems in one dimension
23.	be able to use the equation for the moment of a force, moment of force $= Fx$ where *x* is the perpendicular distance between the line of action of the force and the axis of rotation
24.	be able to use the concept of centre of gravity of an extended body and apply the principle of moments to an extended body in equilibrium
25.	be able to use the equation for work $\Delta W = F\Delta s$, including calculations when the force is not along the line of motion
26.	be able to use the equation $E_k = \frac{1}{2}mv^2$ for the kinetic energy of a body
27.	be able to use the equation $\Delta E_{\text{grav}} = mg\Delta h$ for the difference in gravitational potential energy near the Earth's surface
28.	know, and understand how to apply, the principle of conservation of energy, including use of work done, gravitational potential energy and kinetic energy
29.	be able to use the equations relating power, time and energy transferred or work done $P = \dfrac{E}{t}$ and $P = \dfrac{W}{t}$
30.	be able to use the equations $\text{efficiency} = \dfrac{\text{useful energy output}}{\text{total energy input}}$ and $\text{efficiency} = \dfrac{\text{useful power output}}{\text{total power input}}$

The Sound of Music (MUS)

A study of music and recorded sound, focusing on the production of sound by musical instruments and the operation of a CD/DVD player:

- synthesised and 'natural' sounds
- travelling waves and standing/stationary waves in string and wind instruments
- reading a CD/DVD by laser.

Waves and photons are used to model the behaviour of light.

There are opportunities for students to develop ICT skills and other skills relating to investigation and to communication.

Students should:	
59.	understand the terms *amplitude*, *frequency*, *period*, *speed* and *wavelength*
60.	be able to use the wave equation $v = f\lambda$
61.	be able to describe longitudinal waves in terms of pressure variation and the displacement of molecules
62.	be able to describe transverse waves
63.	be able to draw and interpret graphs representing transverse and longitudinal waves including standing/stationary waves
64.	**CORE PRACTICAL 6: Determine the speed of sound in air using a 2-beam oscilloscope, signal generator, speaker and microphone.**
65.	know and understand what is meant by *wavefront*, *coherence*, *path difference*, *superposition*, *interference* and *phase*
66.	be able to use the relationship between *phase difference* and *path difference*
67.	know what is meant by a *standing/stationary wave* and understand how such a wave is formed, know how to identify nodes and antinodes
68.	be able to use the equation for the speed of a transverse wave on a string $v = \sqrt{\frac{T}{\mu}}$
69.	**CORE PRACTICAL 7: Investigate the effects of length, tension and mass per unit length on the frequency of a vibrating string or wire.**
90.	understand how the behaviour of electromagnetic radiation can be described in terms of a wave model and a photon model, and how these models developed over time
91.	be able to use the equation $E = hf$, that relates the photon energy to the wave frequency
96.	understand atomic line spectra in terms of transitions between discrete energy levels and understand how to calculate the frequency of radiation that could be emitted or absorbed in a transition between energy levels.

Good Enough to Eat (EAT)

A case study of the production of sweets and biscuits:

- measuring and controlling the flow of a viscous liquid
- mechanical testing of products
- using refractometry and polarimetry to monitor sugar concentration.

There are opportunities for students to develop practical techniques and thus to carry out experimental and investigative activities.

Students should:
49. be able to use the equation density $\rho = \frac{m}{V}$
50. understand how to use the relationship upthrust = weight of fluid displaced
51. a. be able to use the equation for viscous drag (Stokes' Law), $F = 6\pi\eta rv$. b. understand that this equation applies only to small spherical objects moving at low speeds with *laminar flow* (or in the absence of *turbulent flow*) and that viscosity is temperature dependent
52. **CORE PRACTICAL 4: Use a falling-ball method to determine the viscosity of a liquid.**
53. be able to use the Hooke's law equation, $\Delta F = k\Delta x$, where k is the stiffness of the object
55. a. be able to draw and interpret force-extension and force-compression graphs b. understand the terms limit of proportionality, *elastic limit, yield point, elastic deformation and plastic deformation* and be able to apply them to these graphs
71. know and understand that at the interface between medium 1 and medium 2 $n_1 sin\theta_1 = n_2 sin\theta_2$ where refractive index is $n = \frac{c}{v}$
72. be able to calculate *critical angle* using $\sin C = \frac{1}{n}$
73. be able to predict whether total internal reflection will occur at an interface
74. understand how to measure the refractive index of a solid material
82. understand what is meant by *plane polarisation*.

Technology in Space (SPC)

The focus is on a satellite whose instruments are run from a solar power supply:

- illuminating solar cells
- operation of photocells
- design and operation of dc circuits
- combining sources of e.m.f.

Mathematical models are developed to describe ohmic behaviours and the variation of resistance with temperature. Simple conceptual models are used for the flow of charge in a circuit, for the operation of a photocell, and for the variation of resistance with temperature.

Waves and photons are used to model the behaviour of light and there is some discussion of the historical development of the photoelectric effect.

There are opportunities to develop ICT skills using the internet, spreadsheets and software for data analysis and display.

Students should:	
31.	understand that electric current is the rate of flow of charged particles and be able to use the equation $I = \frac{\Delta Q}{\Delta t}$
32.	understand how to use the equation $V = \frac{W}{Q}$
33.	understand that resistance is defined by $R = \frac{V}{I}$ and that Ohm's law is a special case when $I \propto V$ for constant temperature
34.	understand how the distribution of current in a circuit is a consequence of charge conservation
35.	understand how the distribution of potential differences in a circuit is a consequence of energy conservation
36.	be able to derive the equations for combining resistances in series and parallel using the principles of charge and energy conservation, and be able to use these equations
37.	be able to use the equations $P = VI$, $W = VIt$ and be able to derive and use related equations, e.g. $P = I^2R$ and $P = \frac{V^2}{R}$
38.	understand how to sketch, recognise and interpret current-potential difference graphs for components, including ohmic conductors, filament bulbs, thermistors and diodes
45.	know the definition of *electromotive force (e.m.f.)* and understand what is meant by *internal resistance* and know how to distinguish between e.m.f. and *terminal potential difference*

Students should:	
47.	understand how changes of resistance with temperature may be modelled in terms of lattice vibrations and number of conduction electrons and understand how to apply this model to metallic conductors and negative temperature coefficient thermistors
48.	understand how changes of resistance with illumination may be modelled in terms of the number of conduction electrons and understand how to apply this model to LDRs
70.	be able to use the equation intensity of radiation $I = \frac{P}{A}$
92.	understand that the absorption of a photon can result in the emission of a photoelectron
93.	understand the terms *threshold frequency* and *work function* and be able to use the equation $hf = \phi + \frac{1}{2}mv_{\max}^2$
94.	be able to use the *electronvolt* (eV) to express small energies
95.	understand how the photoelectric effect provides evidence for the particle nature of electromagnetic radiation.

Digging up the Past (DIG)

The excavation of an archaeological site, from geophysical surveying to artefact analysis:

- resistivity surveying
- artefact analysis by X-ray imaging and diffraction
- artefact analysis by electron microscopy.

Waves are used to model the behaviour of electromagnetic radiation and electrons.

Through a variety of practical and ICT activities, there are opportunities to revisit, review and build on work from previous topics.

Students should:	
39.	be able to use the equation $R = \frac{\rho l}{A}$
40.	**CORE PRACTICAL 2: Determine the electrical resistivity of a material.**
41.	be able to use $I = nqvA$ to explain the large range of resistivities of different materials
42.	understand how the potential along a uniform current-carrying wire varies with the distance along it
43.	understand the principles of a potential divider circuit and understand how to calculate potential differences and resistances in such a circuit
44.	be able to analyse potential divider circuits where one resistance is variable including thermistors and light dependent resistors (LDRs)
83.	understand what is meant by *diffraction* and use Huygens' construction to explain what happens to a wave when it meets a slit or an obstacle
84.	be able to use $n\lambda = d sin\theta$ for a diffraction grating
85.	**CORE PRACTICAL 8: Determine the wavelength of light from a laser or other light source using a diffraction grating.**
86.	understand how diffraction experiments provide evidence for the wave nature of electrons
87.	be able to use the de Broglie equation $\lambda = \frac{h}{p}$

Spare-Part Surgery (SUR)

A study of the physics associated with spare-part surgery for joint replacements and lens implants:

- mechanical properties of bone and replacement materials
- lens implants and the optical system of the eye
- 'designer' materials for medical use
- ultrasound imaging.

Through a variety of practical and ICT activities, there are opportunities to revisit, review and build on work from previous topics.

Students should:	
54.	understand how to use the relationships: • *(tensile or compressive) stress = force/cross-sectional area* • *(tensile or compressive) strain = change in length/original length* • *Young modulus = stress/strain*
56.	be able to draw and interpret tensile or compressive stress-strain graphs, and understand the term *breaking stress*
57.	**CORE PRACTICAL 5: Determine the Young modulus of a material.**
58.	be able to calculate the elastic strain energy E_{el} in a deformed material sample, using the equation $\Delta E_{el} = \frac{1}{2} F \Delta x$, and from the area under the force-extension graph *The estimation of area and hence energy change for both linear and non-linear force-extension graphs is expected*
75.	understand the term *focal length* of converging and diverging lenses
76.	be able to use ray diagrams to trace the path of light through a lens and locate the position of an image
77.	be able to use the equation power of a lens $P = \frac{1}{f}$
78.	understand that for thin lenses in combination $P = P_1 + P_2 + P_3 + \ldots$
79.	know and understand the terms *real image* and *virtual image*
80.	be able to use the equation $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$ for a thin converging or diverging lens with the real-is-positive convention
81.	know and understand that magnification = image height/object height and $m = \frac{v}{u}$
88.	understand that waves can be transmitted and reflected at an interface between media

Transport on Track (TRA)

A study of a modern rail transport system with an emphasis on safety and control:

- track circuits and signalling
- sensing speed
- mechanical braking
- regenerative and eddy current braking
- crash-proofing.

Students use mathematical models to describe the behaviour of moving vehicles and to model electromagnetic induction, capacitor discharge and AC circuits.

There are opportunities to develop ICT skills.

Students should:
97. understand how to use the equation impulse = $F\Delta t = \Delta p$ (Newton's second law of motion)
98. **CORE PRACTICAL 9: Investigate the relationship between the force exerted on an object and its change of momentum.**
101. understand how to determine whether a collision is elastic or inelastic
116. understand that capacitance is defined as $C = \frac{Q}{V}$ and be able to use this equation
118. be able to draw and interpret charge and discharge curves for resistor–capacitor circuits and understand the significance of the time constant *RC*
119. **CORE PRACTICAL 11: Use an oscilloscope or data logger to display and analyse the potential difference (p.d.) across a capacitor as it charges and discharges through a resistor.**
120. be able to use the equation $Q = Q_0 e^{-t/RC}$ and derive and use related equations for exponential discharge in a resistor-capacitor circuit, $I = I_0 e^{-t/RC}$, and $V = V_0 e^{-t/RC}$ and the corresponding log equations $\ln Q = \ln Q_0 - \frac{t}{RC}$, $\ln I = \ln I_0 - \frac{t}{RC}$ and $\ln V = \ln V_0 - \frac{t}{RC}$
121. understand and use the terms *magnetic flux density B*, *flux* ϕ and *flux linkage* $N\phi$
123. be able to use the equation $F = BIl \sin\theta$ and apply Fleming's left-hand rule to current carrying conductors in a magnetic field
124. understand the factors affecting the e.m.f. induced in a coil when there is relative motion between the coil and a permanent magnet
125. understand the factors affecting the e.m.f. induced in a coil when there is a change of current in another coil linked with this coil

Students should:
127. understand how to use Faraday's law to determine the magnitude of an induced e.m.f. and be able to use the equation that combines Faraday's and Lenz's laws $\mathscr{E} = \frac{-\mathrm{d}(N\phi)}{\mathrm{d}t}$
128. understand what is meant by the terms *frequency*, *period*, *peak value* and *root-mean-square value* when applied to alternating currents and potential differences
129. be able to use the equations $V_{rms} = \frac{V_0}{\sqrt{2}}$ and $I_{rms} = \frac{I_0}{\sqrt{2}}$

The Medium is the Message (MDM)

An exploration of the physics involved in a variety of modern communication and display techniques:

- fibre optics and exponential attenuation
- CDD imaging
- cathode ray tube
- liquid crystal and LED displays.

There are opportunities to develop ICT skills and use computer simulations.

Students should:
108. understand that an electric field (force field) is defined as a region where a charged particle experiences a force
109. understand that electric field strength is defined as $E = \frac{F}{Q}$ and be able to use this equation
112. know and understand the relation between electric field and electric potential
113. be able to use the equation $E = \frac{V}{d}$ for an electric field between parallel plates
115. be able to draw and interpret diagrams using field lines and equipotentials to describe radial and uniform electric fields
117. be able to use the equation $W = \frac{1}{2}QV$ for the energy stored by a capacitor, be able to derive the equation from the area under a graph of potential difference against charge stored and be able to derive and use the equations $W = \frac{1}{2}CV^2$ and $W = \frac{\frac{1}{2}Q^2}{C}$
132. understand that electrons are released in the process of thermionic emission and how they can be accelerated by electric and magnetic fields.

Probing the Heart of Matter (PRO)

An area of fundamental physics that is the subject of current research, involving the acceleration and detection of high-energy particles and the interpretation of experiments:

- alpha scattering and the nuclear model of the atom
- accelerating particles to high energies
- detecting and interpreting the interactions between particles
- the quark-lepton model.

Students study the development of the nuclear model and the quark-lepton model to describe matter on a subatomic scale.

There are opportunities to develop ICT skills.

Students should:
99. understand how to apply conservation of linear momentum to problems in two dimensions
100. **CORE PRACTICAL 10: Use ICT to analyse collisions between small spheres, e.g. ball bearings on a table top.**
102. be able to derive and use the equation $E_k = \frac{p^2}{2m}$ for the kinetic energy of a non-relativistic particle
103. be able to express angular displacement in radians and in degrees, and convert between these units
104. understand what is meant by *angular velocity* and be able to use the equations $v = \omega r$ and $T = \frac{2\pi}{\omega}$
105. be able to use vector diagrams to derive the equations for centripetal acceleration $a = \frac{v^2}{r} = r\omega^2$ and understand how to use these equations
106. understand that a resultant force (centripetal force) is required to produce and maintain circular motion
107. be able to use the equations for centripetal force $F = ma = \frac{mv^2}{r} = mr\omega^2$
110. be able to use the equation $F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r^2}$, for the force between two charges
111. be able to use the equation $E = \frac{Q}{4\pi\varepsilon_0 r^2}$ for the electric field due to a point charge

Students should:	
122.	be able to use the equation $F = Bqv \sin\theta$ and apply Fleming's left-hand rule to charged particles moving in a magnetic field
130.	understand what is meant by *nucleon number* (*mass number*) and *proton number* (*atomic number*)
131.	understand how large-angle alpha particle scattering gives evidence for a nuclear model of the atom and how our understanding of atomic structure has changed over time
133.	understand the role of electric and magnetic fields in particle accelerators (linac and cyclotron) and detectors (general principles of ionisation and deflection only)
134.	be able to derive and use the equation $r = \frac{p}{BQ}$ for a charged particle in a magnetic field
135.	be able to apply conservation of charge, energy and momentum to interactions between particles and interpret particle tracks
136.	understand why high energies are required to investigate the structure of nucleons
137.	be able to use the equation $\Delta E = c^2 \Delta m$ in situations involving the creation and annihilation of matter and antimatter particles
138.	be able to use MeV and GeV (energy) and MeV/c^2, GeV/c^2 (mass) and convert between these and SI units
139.	understand situations in which the relativistic increase in particle lifetime is significant (use of relativistic equations not required)
140.	know that in the standard quark-lepton model particles can be classified as: • baryons (e.g. neutrons and protons) which are made from three quarks • mesons (e.g. pions) which are made from a quark and an antiquark • leptons (e.g. electrons and neutrinos) which are fundamental particles • photons and that the symmetry of the model predicted the top quark
141.	know that every particle has a corresponding antiparticle and be able to use the properties of a particle to deduce the properties of its antiparticle and vice versa
142.	understand how to use laws of conservation of charge, baryon number and lepton number to determine whether a particle interaction is possible
143.	be able to write and interpret particle equations given the relevant particle symbols.

Build or Bust? (BLD)

A study of some aspects of building design, including withstanding earthquake damage, vibration isolation and temperature control:

- earthquake detection
- vibration and resonance in structures
- damping vibration using ductile materials
- sensing temperature
- energy, temperature change and phase change.

The behaviour of oscillators is modelled using the mathematics of simple harmonic motion, and physical models are used to explore the behaviour of structures.

Students should:
144. be able to use the equations $\Delta E = mc\Delta\theta$ and $\Delta E = L\Delta m$
145. **CORE PRACTICAL 12: Calibrate a thermistor in a potential divider circuit as a thermostat.**
146. **CORE PRACTICAL 13: Determine the specific latent heat of a phase change.**
181. understand that the condition for simple harmonic motion is $F = -kx$, and hence understand how to identify situations in which simple harmonic motion will occur
182. able to use the equations $a = -\omega^2 x$, $x = A\cos \omega t$, $v = -A\omega \sin \omega t$, $a = -A\omega^2 \cos \omega t$, and $T = \frac{1}{f} = \frac{2\pi}{\omega}$ and $\omega = 2\pi f$ as applied to a simple harmonic oscillator
183. be able to use equations for a simple harmonic oscillator $T = 2\pi\sqrt{\frac{m}{k}}$, and a simple pendulum $T = 2\pi\sqrt{\frac{l}{g}}$
184. be able to draw and interpret a displacement–time graph for an object oscillating and know that the gradient at a point gives the velocity at that point
185. be able to draw and interpret a velocity–time graph for an oscillating object and know that the gradient at a point gives the acceleration at that point
186. understand what is meant by *resonance*
187. **CORE PRACTICAL 16: Determine the value of an unknown mass using the resonant frequencies of the oscillation of known masses.**
188. understand how to apply conservation of energy to damped and undamped oscillating systems
189. understand the distinction between *free* and *forced oscillations*
190. understand how the amplitude of a forced oscillation changes at and around the natural frequency of a system and know, qualitatively, how damping affects resonance

Reach for the Stars (STA)

The focus is on physical interpretation of astronomical observations, the formation and evolution of stars, and the history and future of the universe:

- distances of stars
- masses of stars
- energy sources in stars
- star formation
- star death and the creation of chemical elements
- the history and future of the universe.

Students use the molecular kinetic theory of matter, and the Big Bang model of the universe, and carry out mathematical modelling of gravitational force and radioactive decay.

Students should:
147. understand the concept of *internal energy* as the random distribution of potential and kinetic energy amongst molecules
148. understand the concept of *absolute zero* and how the average kinetic energy of molecules is related to the absolute temperature
149. be able to derive and use the equation, $pV = \frac{1}{3}Nm\langle c^2\rangle$ using the kinetic theory model
150. be able to use the equation $pV = NkT$ for an ideal gas
151. **CORE PRACTICAL 14: Investigate the relationship between pressure and volume of a gas at fixed temperature.**
152. be able to derive and use the equation $\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$
153. understand what is meant by a *black body radiator* and be able to interpret radiation curves for such a radiator
154. be able to use the Stefan-Boltzmann law equation $L = \sigma AT^4$ for black body radiators
155. be able to use Wien's law equation $\lambda_{max}T$ = 2.898 x 10^{-3} m K for black body radiators
156. be able to use the equation $I = \frac{L}{4\pi d^2}$ where L is luminosity and d is distance from the source
157. understand how astronomical distances can be determined using trigonometric parallax
158. understand how astronomical distances can be determined using measurements of intensity received from standard candles (objects of known luminosity)

Students should:	
161.	understand how the movement of a source of waves relative to an observer/detector gives rise to a shift in frequency (Doppler effect)
162.	be able to use the equations for redshift $z = \frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$ for a source of electromagnetic radiation moving relative to an observer and $v = H_0 d$ for objects at cosmological distances
163.	understand the controversy over the age and ultimate fate of the universe associated with the value of the Hubble constant and the possible existence of dark matter
164.	understand the concept of *nuclear binding energy*, and be able to use the equation $\Delta E = c^2 \Delta m$ in calculations of nuclear mass (including *mass deficit*) and energy
165.	use the *atomic mass unit (u)* to express small masses, and convert between this and SI units
166.	understand the processes of nuclear fusion and fission with reference to the binding energy per nucleon curve
167.	understand the mechanism of nuclear fusion and the need for very high densities of matter and very high temperatures to bring about and maintain nuclear fusion
168.	understand that there is background radiation and how to take appropriate account of it in calculations
169.	understand the relationships between the nature, penetration, ionising ability and range in different materials of nuclear radiations (alpha, beta and gamma)
170.	be able to write and interpret nuclear equations given the relevant particle symbols
171.	**CORE PRACTICAL 15: Investigate the absorption of gamma radiation by lead.**
172.	understand the spontaneous and random nature of nuclear decay
173.	be able to determine the half-lives of radioactive isotopes graphically and be able to use the equations for radioactive decay: activity $A = \lambda N$, $\frac{dN}{dt} = -\lambda N$, $\lambda = \frac{\ln 2}{t_{1/2}}$, $N = N_0 e^{-\lambda t}$ and $A = A_0 e^{-\lambda t}$ and derive and use the corresponding log equations.
174.	understand that a gravitational field (force field) is defined as a region where a mass experiences a force
175.	understand that gravitational field strength is defined as $g = \frac{F}{m}$ and be able to use this equation

Students should:
177. be able to derive and use the equation $g = \frac{Gm}{r^2}$ for the gravitational field due to a point mass
178. be able to use the equation $V_{grav} = \frac{-Gm}{r}$ for a radial gravitational field
179. be able to compare electric fields with gravitational fields
180. be able to apply Newton's laws of motion and universal gravitation to orbital motion.

Science Practical Endorsement

Overview

The assessment of practical skills is a compulsory requirement of the course of study for A level qualifications in biology, chemistry and physics. It will appear on all students' certificates as a separately reported result, alongside the overall grade for the qualification. The arrangements for the assessment of practical skills will be common to all awarding organisations. These arrangements include:

- A minimum of 12 practical activities to be carried out by each student which, together, meet the requirements of Appendices 5b (Practical skills identified for direct assessment and developed through teaching and learning) and 5c (Use of apparatus and techniques) from the prescribed subject content, published by the Department for Education. The required practical activities will be defined by each awarding organisation.
- Teachers will assess students against Common Practical Assessment Criteria (CPAC) issued by the awarding organisations. The CPAC are based on the requirements of *Appendices 5b* and *5c* of the subject content requirements published by the Department for Education, and define the minimum standard required for the achievement of a pass.
- Each student will keep an appropriate record of their practical work, including their assessed practical activities
- Students who demonstrate the required standard across all the requirements of the CPAC will receive a 'pass' grade
- There will be no separate assessment of practical skills for AS qualifications.
- Students will answer questions in the AS and A level examination papers that assess the requirements of *Appendix 5a* (Practical skills identified for indirect assessment and developed through teaching and learning) from the prescribed subject content, published by the Department for Education. These questions may draw on, or range beyond, the practical activities included in the specification.

Setting practical work

Teaching and learning

Teachers should ensure that the core practicals listed in the subject content are incorporated into teaching and learning and that students carry them out. This is to support development of competency for the practical endorsement but also because students will be indirectly assessed on their practical skills in the examinations. Teachers should consider setting additional practical work to support teaching and learning.

Teachers must devise and retain a teaching plan that shows when practicals will be covered in teaching and learning.

Conditions for assessing practical work

Authenticity

Students and teachers must sign the Practical competency authentication sheet (see *Appendix 4*).

Collaboration

Students may work in pairs on some practicals, where this is appropriate, provided they are able to produce individual evidence to meet the competency statements.

Feedback

Teachers may help students to understand instructions and may provide feedback to support development of the competencies. Teachers should be confident that students have developed their skills and thus students should be able to demonstrate the competencies independently by the end of the course.

Resources

Students must have access to the equipment needed to carry out the core practicals. Students must also have access to IT and internet facilities.

Supervision

Teachers must ensure that students are supervised appropriately.

Evidence of practical work

Evidence should be collected of practical work that is sufficient to show that the competencies have been achieved. Evidence may take a variety of forms.

Students should attempt all 16 practicals given in the qualification content.

Observations and notes by teachers

Teachers should observe sufficient student practicals to ensure that competencies have been achieved.

Student records

Students should keep notes of their practical work sufficient to show evidence of the practical competencies on making and recording observations and researching, referencing and reporting.

Assessing practical work

Teachers should make a judgement of student practical competence using the Common Practical Assessment Criteria on the following pages. Teachers should include any comments on the Practical competency authentication sheet (see *Appendix 4*) to justify the decision made.

The practical activities prescribed in the specification provide opportunities for demonstrating competence in all the skills identified, together with the use of apparatus and techniques for each subject. However, students can also demonstrate these competencies in any additional practical activity undertaken throughout the course of study which covers the requirements of *Appendix 5c*.

Common Practical Assessment Criteria (CPAC)

Teachers must assess student practicals against the following competencies.

Criteria for the assessment of GCE Science practical competency for Biology, Chemistry and Physics	
Competency	**The criteria for a pass** In order to be awarded a Pass a learner must, by the end of the practical science assessment, consistently and routinely meet the criteria in respect of each competency listed below. A learner may demonstrate the competencies in any practical activity undertaken as part of that assessment throughout the course of study. Learners may undertake practical activities in groups. However, the evidence generated by each learner must demonstrate that he or she independently meets the criteria outlined below in respect of each competency. Such evidence: (a) will comprise of both the learner's performance during each practical activity and his or her contemporaneous record of the work that he or she has undertaken during that activity, and (b) must include evidence of independent application of investigative approaches and methods to practical work.
1. Follows written procedures	a) Correctly follows written instructions to carry out the experimental techniques or procedures.
2. Applies investigative approaches and methods when using instruments and equipment	a) Correctly uses appropriate instrumentation, apparatus and materials (including ICT) to carry out investigative activities, experimental techniques and procedures with minimal assistance or prompting. b) Carries out techniques or procedures methodically, in sequence and in combination, identifying practical issues and making adjustments when necessary. c) Identifies and controls significant quantitative variables where applicable, and plans approaches to take account of variables that cannot readily be controlled. d) Selects appropriate equipment and measurement strategies in order to ensure suitably accurate results.

Criteria for the assessment of GCE Science practical competency for Biology, Chemistry and Physics	
3. Safely uses a range of practical equipment and materials	a) Identifies hazards and assesses risks associated with these hazards, making safety adjustments as necessary, when carrying out experimental techniques and procedures in the lab or field. b) Uses appropriate safety equipment and approaches to minimise risks with minimal prompting.
4. Makes and records observations	a) Makes accurate observations relevant to the experimental or investigative procedure. b) Obtains accurate, precise and sufficient data for experimental and investigative procedures and records this methodically using appropriate units and conventions.
5. Researches, references and reports	a) Uses appropriate software and/or tools to process data, carry out research and report findings. b) Cites sources of information demonstrating that research has taken place, supporting planning and conclusions.

Marking and standardisation

The practical work is assessed by teachers. Pearson will support teachers in making judgements against the criteria for assessment.

In coordination with other exam boards, Pearson will monitor how schools provide students with opportunities to develop and demonstrate the required practical skills and how they mark the assessments.

Every school will be monitored at least once in a two-year period in respect of at least one of the A level science subjects. These monitoring visits will be coordinated by JCQ, who will undertake communications with centres to facilitate the allocation of exam board monitoring visits.

In common with other exam boards, Pearson will require centres to provide a statement confirming they have taken reasonable steps to secure that students:

- undertook the minimum number of practical activities, and

Students will only get a certificate for the practical assessment if they achieve at least a grade E in the examined part of the qualification.

Students who do not pass the practical assessment will have a 'Not Classified' outcome included on their certificate unless they were exempt from the assessment because of a disability.

Malpractice

Candidate malpractice

Candidate malpractice refers to any act by a candidate that compromises or seeks to compromise the process of assessment or which undermines the integrity of the qualifications or the validity of results/certificates.

Candidate malpractice in controlled assessments discovered before the candidate has signed the declaration of authentication form does not need to be reported to Pearson.

Candidate malpractice found in controlled assessments after the declaration of authenticity has been signed, and in examinations **must** be reported to Pearson on a *JCQ Form M1* (available at www.jcq.org.uk/exams-office/malpractice). The completed form can be emailed to pqsmalpractice@pearson.com or posted to Investigations Team, Pearson, 190 High Holborn, London, WC1V 7BH. Please provide as much information and supporting documentation as possible. Note that the final decision regarding appropriate sanctions lies with Pearson.

Failure to report candidate malpractice constitutes staff or centre malpractice.

Staff/centre malpractice

Staff and centre malpractice includes both deliberate malpractice and maladministration of our qualifications. As with candidate malpractice, staff and centre malpractice is any act that compromises or seeks to compromise the process of assessment or undermines the integrity of the qualifications or the validity of results/certificates.

All cases of suspected staff malpractice and maladministration **must** be reported immediately, before any investigation is undertaken by the centre, to *Pearson on a JCQ Form M2(a)* (available at www.jcq.org.uk/exams-office/malpractice). The form, supporting documentation and as much information as possible can be emailed to pqsmalpractice@pearson.com or posted to Investigations Team, Pearson, 190 High Holborn, London, WC1V 7BH. Note that the final decision regarding appropriate sanctions lies with Pearson.

Failure to report malpractice itself constitutes malpractice.

More detailed guidance on malpractice can be found in the latest version of the document *General and Vocational Qualifications Suspected Malpractice in Examinations and Assessments Policies and Procedures,* available at www.jcq.org.uk/exams-office/malpractice.

Assessment

Assessment summary

Summary of table of assessment

Students must complete all assessment in May/June in any single year.